MIL-W-81319A(AS)
22 August 1967
Superseding
MIL-W-81319(WP)
8 September 1965

MILITARY SPECIFICATION

WHEELS, ABRASIVE: NON-WOVEN NYLON, RESIN REINFORCED

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for reinforced abrasive wheels consisting of abrasive particles, non-woven nylon fibers and a thermosetting resin binder. (see 6.1).

1.2 Classification - The abrasive wheels shall be of the following Classes, Types and Grades as specified. (See 3.3 for availability of abrasive wheels in each classification)

Classes

Class 1	Convolutely wound
Class 2	Laminated

Types

Type I	Aluminum oxide (Al_2O_3)
Type II	Flint (SiO_2)
Type III	Silicon carbide (SiC)

* Grades

AAA	Ultrafine
AA	Superfine
A	Very fine
B	Fine
C	Medium

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

FSC 5345

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SPECIFICATIONS

Federal

QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet
TT-T-291	Thinner, Paint, Volatile Mineral Spirits (Petroleum Spirits)
VV-O-251	Oil, Cutting, Mineral-Fatty-Oil Blend

Military

MIL-P-3816	Abrasives and Abrasive Products for Shipment and Storage, Packaging and Packing of
MIL-C-22543	Cleaning Compound, Water Emulsion
MIL-C-22550	Cleaning Compound, Aircraft Surface, Waterless
MIL-L-23699	Lubricating Oil, Aircraft Turboprop and Turboshaft Engines, Synthetic Base

STANDARDS

Federal

FED. TEST METHOD STD. NO. 141	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-120	Gage Inspection
MIL-STD-129	Marking for Shipment and Storage

(When requesting specifications, standards and publications refer to both the title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot (105), 5801 Tabor Avenue, Philadelphia, Penna., 19120. All other documents should be obtained from the procuring activity, or as directed by the Contracting Officer.)

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2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

Department of Commerce

Commercial Standard:

CS217 - - - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Data - No data is required by this specification or by the applicable documents referenced in Section 2 unless specified in the contract or order. (See 6.2)

3.2 Materials - The abrasive wheels shall consist of abrasive particles, non-woven nylon fibers and a thermosetting resin binder. The manufacturer shall certify in writing that the abrasive wheels to be supplied conform to the requirements specified in 3.2.1, 3.2.1.1, 3.2.2, and 3.2.3 (see 4.4.1).

* 3.2.1 Abrasive particles

Type I - Type I abrasive shall consist of no less than 94 percent aluminum oxide (Al_2O_3). The hardness of the aluminum oxide shall be 9.4 Moh's scale.

Type II - Type II abrasive shall consist of not less than 98 percent of a mild, natural, high quality Flint (SiO_2). The hardness of the Flint shall be not less than 6.8 nor more than 7.0 Moh's scale.

Type III - Type III abrasive shall consist of not less than 97 percent silicon carbide (SiC). The hardness of the silicon carbide shall be 9.6 Moh's scale.

* 3.2.1.1 Abrasive grit size - When graded in accordance with the U.S. Department of Commerce Standard CS 217-59, the abrasive shall conform to the requirements of Table I. The abrasive grit requirements specified in Table I shall apply to abrasive particles that have been recovered from the abrasive wheel.

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Table I

Grit size (microns)

Grades	Average	Tolerance	
		% over	% under
AAA (Ultrafine)	less than 10 1/	-	-
AA (Superfine)	18 ± 5	25	10
A (Very fine)	38 ± 5	35	55
B (Fine)	95 ± 5 2/	-	-
C (Medium)	150 ± 10 3/	-	-

1/ no particles greater than 35 microns

2/ no particles greater than 130 microns, or less than 25 microns

3/ no particles greater than 330 microns or less than 50 microns.

* 3.2.2 Fiber - Unless otherwise specified by the Procuring Activity, the fiber used in the abrasive wheel shall be a nylon filament conforming to the requirements of Table II.

Table II

Property	Grades AAA, AA, A, and B	Grade C
Material	Nylon	Nylon
Polymer Type	Type 6 or Type 6-6	Type 6 or Type 6-6
Denier	15 ± 5	58 ± 10
Crimps per inch	12 ± 5	12 ± 5
Heat set	Yes	Yes
Staple length	1 1/2 ± 1/8 inch	2 ± 3/16 inch
Tenacity - grams		
per denier	3.0 to 6.0	2.5 to 4.5
% elongation	40 to 140	50 to 150

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3.2.3 Resin binder - The binder used in the abrasive wheel shall be a good quality thermosetting resin.

3.3 Availability of abrasive wheels in each classification - Unless otherwise specified by the Procuring Activity, abrasive wheels shall be furnished in accordance with Table III.

Table III

Classes	Type	Grade
1 and 2	I (Al_2O_3)	A, B, and C
1 and 2	II (Flint)	B only
1 and 2	III (SiC)	AAA, AA, A, B and C

* 3.4 Abrasive wheel dimensions - Unless otherwise specified by the Procuring Activity, the available Grades specified in 3.3 shall be furnished in the sizes shown in Table IV.

Table IV a/

Class	O.D.	I.D.	Width c/	Maximum Safe R.P.M. b/
1	20	12	1 to 7	1100
1	17	10	1 to 7	1500
1	16	10	1 to 7	1500
1	14	8	1 to 7	2000
1	12	5	1 to 7	2500
1	8	3	1 to 7	3200
1	6	1	1 to 7	4500
1	4	1	1 to 7	4500
2	6	1	1/4, 1/2, 3/4, 1	4800
2	3	1/2	1/4, 1/2, 3/4, 1	8000
2	2	1/4	1/4, 1/2	11,000
2	1 1/2	1/8	1/4	13,000

a/ all dimensions in inches
b/ When specified in the Contract or Order, the maximum safe RPM shall be subject to the approval of the Procuring Activity.(see 6.2)
c/ in increments of one inch

3.4.1 Tolerances - Unless otherwise specified by the Procuring Activity, dimensional tolerances shall be in accordance with Table V and 3.4.1.1.

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Table V

Dimension	Tolerance
O. D. (up to 16 inches)	± 1/16 inch
O. D. (17 to 20 inches)	± 1/8 inch
I. D. (all sizes)	± 0.010 inch
width (up to 1 inch)	+ 1/16 - 0 inch
width (2 to 7 inches)	± 1/16 inch

* 3.4.1.1 Concentricity (all sizes) - When concentricity is measured as specified in paragraph 8.4.4 of MIL-STD-120, the total indicator runout (T.I.R.) shall not exceed 0.030 inch.

* 3.5 Maximum safe R. P. M. - When tested as specified in 4.5.1, the abrasive wheel shall show no cracking, loss of material or other deterioration.

3.6 Corrosiveness - When subjected to the salt spray test specified in 4.5.2, the polished area of each aluminum alloy panel shall show no more corrosive attack than the unpolished area of each panel.

* 3.7 Fluid effects - When tested with water and the various non-acidic fluids as specified in 4.5.3, the abrasive wheel shall show no sign of softening, cracking, chemical attack, or any other deterioration.

3.8 Performance - During the test specified in 4.5.2, the abrasive wheel shall cut at an even rate and show an even wear pattern. The abrasive wheel shall be clean running during the polishing operation; and there shall be no adherent residue on the aluminum alloy panels after the polishing operation.

* 3.9 Identification - Each abrasive wheel shall be marked in accordance with MIL-STD-129 with the following:

Manufacturer's designation
Type of abrasive
Abrasive Grade
Density
Maximum safe operating speed (RPM)

3.10 Workmanship - When visually examined, the abrasive wheel shall be uniform in the distribution of its ingredient materials. The wheel shall be free from unevenness, cracks, voids, lumps, foreign material, excessively large abrasive particles or any other defect that would be detrimental to appearance, fabrication or performance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot formation - A lot shall consist of not more than 1500 abrasive wheels of one Class, Type and Grade that have been manufactured under the same manufacturing process, at essentially the same time and presented for inspection at one time.

4.3 Quality conformance samples -

* 4.3.1 Samples for tests - The abrasive wheels shall be selected at random from each lot in accordance with Table VII.

* 4.3.2 Samples of the end item and preparation for delivery - Samples for the inspections specified in 4.4.2 shall be selected at random from each lot in accordance with MIL-STD-105 and Table VI. The lot size shall be the number of abrasive wheels for inspection 4.4.2.1, the number of filled unit containers for 4.4.2.2, the number of filled intermediate containers for 4.4.2.3, and the number of shipping containers for 4.4.2.4.

4.4 Quality conformance inspection -

4.4.1 Inspection of samples for tests - The samples selected in accordance with Table VII shall be examined and tested to determine conformance to requirements specified in Table VII. The manufacturer shall certify in writing that the abrasive wheels to be supplied conform to these requirements:

3.2.1 - Abrasive; 3.2.1.1 - Grit size; 3.2.2 - Fiber; 3.2.3 - Resin binder. Non-conformance of a sample to a single requirement shall be cause for rejection of the lot represented by the sample.

* 4.4.2 Inspection of the end item and preparation for delivery - The applicable samples selected in accordance with 4.3.2 and Table VI shall be examined for the defects classified in 4.4.2.1, 4.4.2.2, 4.4.2.3 and 4.4.2.4 and to determine compliance with all the other applicable requirements of Section 5 of this specification. If the number of defective samples exceeds the acceptable quality level specified in Table VI, the lot represented by the sample shall be rejected.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-W-81319A(AS) WHEELS, ABRASIVE, NON-WOVEN NYLON, RESIN REINFORCED

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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